

Understanding and Reducing Perception Gaps with Mediated Social Cues when Building Workplace Relationships through CMC

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ABSTRACT

More and more people form and maintain workplace relationships through computer-mediated communication tools nowadays. When all interactions are mediated by technology, people's impression of each other can be affected by the selective social cues transmitted by media. The way people present themselves through communication technology might not be perceived by their remote counterparts in the same way as they do. The mediated social cues could affect how people interpret the perceived action, further affecting impression formation and relationship maintenance between remote collaborators. It is unclear whether and how such mediated social cues cause perception gaps and how such perception gaps among different social connections play a part in relationship building in remote workplaces. The goal of my doctoral thesis is to identify the perception gap in computer-mediated communication and its impact on workplace relationship building. First, in a mixed-methods study in remote workplaces, I found that people's perceptions of their strongly and weakly connected colleagues become polarized when their collaboration is mediated by different communication technologies. Next, I specifically focused on relationship building between potential collaborators, which is another type of weakly connected collaborators in workplaces, via videoconferencing. In a controlled online study, I found that ambient cues from virtual backgrounds cause different impacts on first impressions of remote collaborators and perception of self. With this doctoral thesis, I aim to provide design implications on designing social cues to reduce the perception gaps introduced by communication technology in different stages of workplace relationship building.

KEYWORDS

Perception gap, social cues, computer-mediated communication, misattribution, workplace relationship

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1 INTRODUCTION AND BACKGROUND

Impression management plays an important role in affecting whether people want to connect and maintain collaboration further with each other, and it is especially critical in building relationships in virtual teams. Increasingly, people form and maintain workplace relationships through computer-mediated communication tools nowadays, especially under the influence of pandemics. Computer-mediated communication tools enable people to initiate collaboration and build relationships with potential collaborators without being affected by spatial constraints. During social interactions, nonverbal cues provide rich information, such as tone, head movement frequency, or duration of eye contact [2] for people to judge one's behavior and personality. However, when all interactions are moved online, the lack of nonverbal information makes it harder for people to build relationships. For instance, it has been found that compared to face-to-face interaction, people take longer time to build mutual trust via text messaging [4, 22, 23]. Moreover, when all social interactions are mediated by technology, people's impressions and perception of each other can be affected by the mediated social cues transmitted by different media. For example, the profile images people used, the description of one's self-introduction, and visible online activities can influence how people perceive one another and their willingness to form further relationships with each other to various extents [6, 17, 20].

When social interactions are mediated by communication technologies, there are spaces for people to present themselves with selective social cues. In videoconferencing, people could adjust their field of view or angle of the camera to monitor whether they appeared agreeable in front of the video [15]. In addition, people can select ideal profile images by using filters [5] or by setting up virtual backgrounds to present themselves in a preferable way in front of their remote counterparts. Their remote counterparts, who receive such mediated social cues, can form various impressions based on the mediated social cues, and may further affect how they interpret the perceived action, either positively or negatively. For instance, study showed that talking to a remote partner by seeing him/her being embodied in an avatar with an enhanced smile could increase the positive impression during the remote conversation [18]. On the other hand, it was found that remote workers interpreted remote colleagues as less engaged when seeing their camera off in videoconferencing, compared to other attendees who turn on the camera [11]. These studies suggested that there might be perception gaps in computer-mediated communication, and it can

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affect relationship building when the mediated social cues are not perceived as intended.

When people present themselves through communication technologies, they might not be perceived by their remote interlocutors in the same way as they expected. Such perception gap between the sender and the receiver can influence the way people interpret the observed action from each other, which might lead to misunderstandings about each other's intentions and judgment of the situation. The lack of nonverbal cues in computer-mediated communication can further enlarge the perception gap between the sender and the receiver, and misinterpretation might occur. For example, study showed that when people conducted discussions with cross-lingual members, native speakers could misattribute non-native speakers' behavior by observing nonverbal cues through videoconferencing [8]. Moreover, it has been identified that when people are explaining the observed behavior they received from communication technologies, their initial impressions of the remote counterparts become polarized, which is called overattribution [21]. The insufficient social cues lead people to exacerbate the positive and negative impression when their interactions are carried out over communication technologies, which might increase the perception gap and its impact on relationship building.

In workplaces, workers rely on many social cues to infer one another's engagement level in work, and such perceptions could impact workers' social relations and the outcome of their collaboration [7, 14]. When working remotely, communication technologies and online collaboration tools are the main source for people to interact with and make inferences about each other. Despite the design of various computer-mediated communication and team communication tools to make remote collaboration more effective, such as supporting awareness-checking [3, 16] and using telepresence robots to initiate informal conversations across distance [12, 13], remote workers still face social isolation [10], and have difficulties in maintaining social relationships with their remote colleagues [1]. When all social interactions are mediated by technology, it becomes unclear how workers access the mediated social cues to (1) form relationships with first encounters, who are potential collaborators but lack of basic mutual understanding, and (2) maintain existing relationships in various types of social connections, such as co-workers who meet five times a week and co-workers who have only communicated once in the last month.

Therefore, in this thesis, I aim to investigate whether and how computer-mediated social cues affect people's perceptions of each other, thus further influencing their relationship building in remote workplaces. Specifically, I ask the following research questions:

RQ1: Will communication technologies generate perception gaps when people build workplace relationships with each other remotely?

RQ2: How do mediated social cues influence perception gaps among people with different social connections?

RQ3: How do different perception gaps affect relationship building in remote workplaces?

2 WORK IN PROGRESS AND EXPECTED CONTRIBUTION

2.1 Study 1: Distance Matters to Weak Ties in Remote Workplaces

I first started out by understanding how people maintain relationships with their remote colleagues with strong and weak social connections when all social interactions are mediated by communication technologies in remote workplaces. I conducted a mixed-methods study with a survey, five-day experience sampling, and an interview with 20 workers who switched their collaboration from collocated to remote due to the pandemic in 2020. I asked workers to report what communication technologies they used to communicate with one of their strongly connected colleagues (whom I defined as strong tie), and one of their weakly connected colleagues (weak tie). Then, all workers were asked to make inferences about their remote colleagues' engagement level and indicate their willingness to collaborate with respective ties three times when working from home via experience sampling. In the final interview, I further explored what cues did workers rely on to make inferences to their remote colleagues and identify what challenges they faced when maintaining awareness about their remote colleagues when working from home.

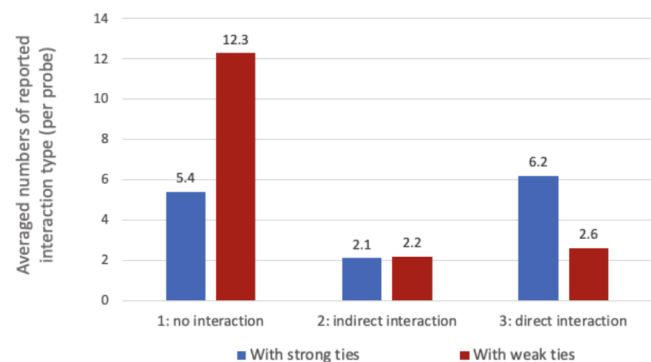


Figure 1: Types of interactions workers had with each tie. The y-axis is the averaged frequency of each type of interaction reported by all workers through the experience sampling method. The x-axis shows three different types of interactions, *no interaction*, *indirect interaction*, *direct interaction*. The bars in blue indicate workers' interactions with their strong ties, while the red bars indicate workers' interactions with their weak ties.

Results showed that workers' perceptions of their strongly and weakly connected colleagues became polarized when all interactions were mediated by communication technologies. Contrary to the frequent and one-on-one interactions with their strong ties, the infrequent and group-based interaction with their weak ties combined with insufficient social cues received in remote workplaces made workers perceive their weak ties as less engaged and showed less willingness to collaborate (Fig. 2). Frequent and one-on-one interactions among strong ties could compensate for the insufficient awareness in remote workplaces, so workers could still maintain

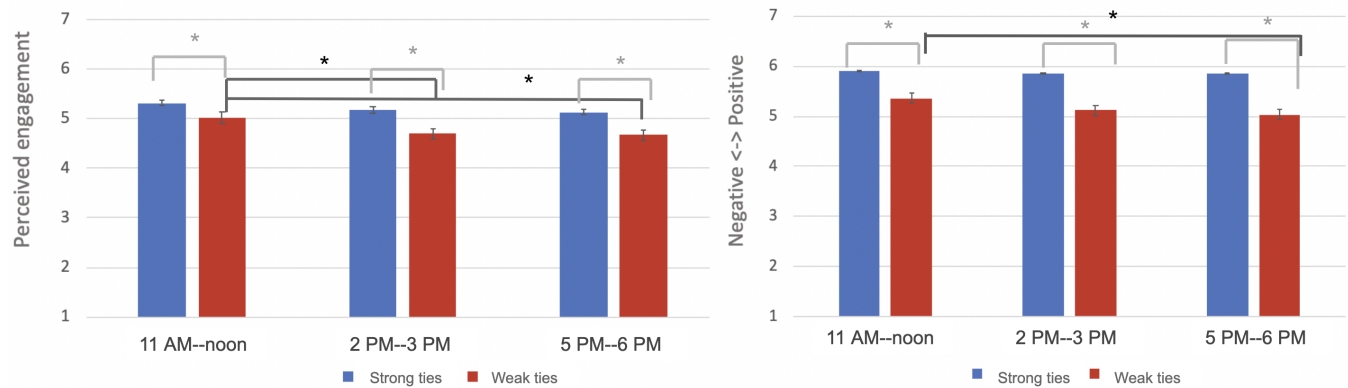


Figure 2: Workers’ perceived engagement and willingness to collaborate with their selected strong and weak ties. Left: Perceived engagement. The y-axis shows the level of perceived engagement by worker. The higher the score, the higher the perceived engagement. Workers perceived weak ties as less engaged than strong ties. **Right: Willingness to collaborate.** The y-axis shows how positively the worker feels about forming a collaboration with each tie. 7 is very positive, while 1 is very negative. Workers showed less willingness to collaborate with weak ties than strong ties.

mutual understanding with each other. However, the infrequent and group-based discussion prevents weak ties to have spontaneous conversations (Fig. 1). This, combined with insufficient awareness in remote workplaces, left a large space for workers to interpret the social cues received from diverse communication technologies. When there were no cues for workers to make inferences about their weak ties, prior impressions formed in the office dominated the inference, which could lead to either extremely positive or negative interpretations of the observed behavior. Interview findings further revealed that workers not only relied on available cues (e.g., social cues received from direct interactions, digital traces, etc) to make inferences about their remote colleagues, but also the unavailable cues. When there was no interaction and other sources in the online workspace for workers to update their impression, the impression formed in prior interactions would dominate the assessment. The finding highlights the importance of enhancing awareness and supporting impression management for weakly connected colleagues, such as newcomers and potential collaborators, in remote workplaces.

2.2 Study 2: Ambient Influences on Impression Formation of First Encounters in Videoconferencing

Study 1 pointed out that there were perception gaps when people interpreted mediated social cues from different social connections, and weakly connected colleagues can be affected by remote work more than strongly connected ones. In study 2, I moved further to examine one type of weak ties, who are potential collaborators that without prior interaction and their first encounter with each other is through videoconferencing. Previous studies have proposed to change one’s appearance by adding filters, using virtual avatars [9], or directly changing facial expressions [19] for better self-presentation in videoconferencing. Different from changing the appearance of people, I proposed to use ambient cues to change remote dyads’ impression of each other. I examined how ambient

cues influence impression formation in videoconferencing, and conducted a controlled online study with 29 dyads. One of each dyad (Role A) was asked to change their virtual background into the one that they perceive themselves to be the most and least professional depending on the experimental conditions they were assigned to. For example, a virtual background with a bookshelf can be selected as having the most professional image or a virtual background with a random illustration can be selected as having the least professional image. Their partners (Role B) were asked to use a gray virtual background as a control comparison. Dyads performed a persuasion task through videoconferencing and rated their perceptions about themselves and their remote counterparts after discussion.

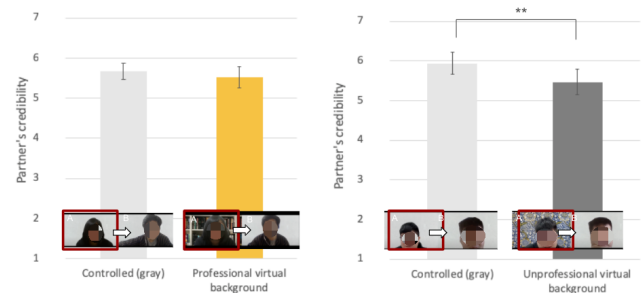


Figure 3: Assessing counterpart’s credibility from Role A participant’s viewpoints. X-axis indicates the controlled gray condition and professional (left) / unprofessional (right) background condition. Y-axis shows the perceived counterpart’s credibility. Changing one’s own virtual backgrounds to unprofessional ones made people perceive remote counterparts as less credible, compared with the controlled gray condition.

Results showed that people's own virtual background, rather than their counterparts' virtual background posed impacts on assessing their remote counterpart's credibility and agreement level of discussion. When using a virtual background that made oneself feel unprofessional caused participants evaluate their counterparts as less credible than both using the same controlled virtual background (Fig. 3). On the other hand, using a virtual background that made oneself feel professional caused participants feel both of them had less agreement on the consensus than the controlled virtual background. This study revealed that ambient cues of the virtual background lead to perception gaps between self and others, which affect their judgement of their remote acquaintances and the agreement in the discussion.

In Study 3, I will focus on supporting another type of weak tie, which is remote newcomers who had no prior connections with existing workers, to build relationships with remote colleagues via text messaging. In this study, I aim to investigate whether there are also perception gaps when people interpret remote newcomers' text messages when receiving enhanced social cues through an avatar.

With three consecutive studies, I aim to: (1) reveal that perception gaps can be manifested and enlarged when the interaction is mediated by communication technologies, and (2) to provide implications for designing social cues to change and reduce perception gaps when building workplace relationships by communication technologies.

3 MOTIVATION TO ATTEND CSCW DC

I aim to have my doctoral thesis defense within a year and a half. By participating CSCW DC, I hope to seek feedback from other peers and senior CSCW researchers about the current framing and the intended research contribution of my research. It would be helpful if I can collect some questions from the attendees so that I can better improve my design decision and rationale for the study. In addition to the research discussion, I also expect to gain experience in mentoring each other during this event. After getting my doctoral degree, I aim to continue doing research as a researcher in academia. I think DC will be a great place for me to seek advice on how to mentor other researchers and how to manage my career trajectories as a female CSCW researcher in Asia.

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